B.Sc. (Fifth Semester) (NEP)

Examination, 2024-25

MATHEMATICS

Paper - II (Theory)

[Linear Algebra]

Time : 3 Hours] **[Maximum Marks : 75**

Note : This question paper contains *two* sections A and B. Attempt any *five* questions each of 6 marks from Section A and any *three* questions each 15 marks from Section B.

SECTION–A

(5×6=30)

1. Let S be a non empty set and F a field. Prove that for any $s_0 \in S$, $\{f \in \mathcal{F}(S, F) : f(s_0) = 0\}$, is a subspace of $\mathcal{F}(S, F)$.

2. Let $S = \{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ be a subset of the vector space $\mathbb{R}^3$. Then show that S is linearly independent.

3. Show that the set $S = \{(1, 0, -1), (2, 5, 1), (0, -4, 3)\}$ is basic for R^3.

4. If $T : R^2 \to R^3$ defined by

$$T(a_1, a_2) = (a_1 + a_2, 0, 2a_1 - a_2)$$

Then prove that T is a linear Transformation.

5. Let T be a linear operator on V, such that $V = R^2$ and $T(a, b) = (-2a + 3b, -10a + 9b)$. Find the eigen values of T and an ordered bases β for V such that $[T]\beta$ is a diagromal matrix.

6. Explain invariant subspaces.

7. Let V be an inner product space over F. Then all $x, y, \in V$,

$$|<x, y>| < \|x\|.\|y\|.$$

8. Define :

(a) Diagonalizable Operator.

(b) Annihilators.

SECTION–B

(3×15=45)

9. Let w_1 and w_2 be subspaces of a vector space V, then

(a) Prove that $w_1 + w_2$ is a subspace of V that contains both w_1 and w_2.

(b) Prove that any subspace of V that contains both w_1 and w_2 must also contain $w_1 + w_2$.

10. Let V and W be vector space, and let $T : V \to W$ be linear. If V is finite dimensional, then

$$\text{nullity}(T) + \text{rank}(T) = \dim(V).$$

11. State and prove the Cayley-Hamilton Theorem.

12. Prove that the restriction of a linear operator T to a T-invariant subspace is a linear operator on that subspace.

13. Apply the Gram-Schmidt process to the given subset S of the inner product space V to obtain an orthogonal basis for span (S), where $V = R^3$, $S = \{(1, 0, 1), (0, 1, 1), (1, 3, 3)\}$.